



Department of Computer Technology

Vision of the Department

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering: Dynamic learning environment that encourages problem-solving using emerging technology, · National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration, Holistic, ethical and sustainable development for lifelong success.

Session 2026-2027(ODD)

Vision: Dream of where you want.	Mission: Means to achieve Vision
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Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date
(Signature and Date in Handwritten)



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Session	2026-27 (ODD)	Course Name	LAB: Software Engineering
Semester	7th	Course Code	23CT1704
Roll No	25	Name of Student	Tanushri Sawalekar

Practical Number	02
Aim	To study and create Software Requirement Specification document for the given case study
Theory (100 words)	SRS for E-Learning Platform In today's digital education environment, the efficiency, accessibility, and reliability of e-learning platforms are essential. Crafting a well-defined Software Requirement Specification (SRS) for an e-learning platform ensures that the system meets all necessary standards, functionalities, and user expectations. This document explores the essential components, benefits, and steps involved in developing a comprehensive SRS for an e-learning platform. The SRS serves as a critical document that outlines the requirements and functionalities of the system. This specification acts as a roadmap for developers, stakeholders, instructors, and learners, ensuring that the final product meets all educational and technical requirements. By detailing every aspect of the system, from user interface design to backend processes, an effective SRS helps avoid costly errors and project delays. Table of Contents • Importance of SRS for E-Learning Platform • Components of an SRS for E-Learning Platform 1. System Overview 2. Functional Requirements 3. Non-Functional Requirements 4. User Interfaces 5. System Architecture 6. Data Requirements 7. External Interface Requirements 8. System Constraints • Developing an SRS for E-Learning Platform



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	1. Requirement Gathering 2. Requirement Analysis 3. Requirement Specification 4. Requirement Validation 5. Requirement Management • Challenges in Developing an SRS for E-Learning Platform 1. Diverse Learning Requirements 2. Data Privacy and Security 3. Integration with Third-Party Services 4. User Adoption and Engagement • Benefits of a Well-Defined SRS for E-Learning Platform 1. Improved Communication 2. Enhanced Project Management 3. Better Quality Assurance 4. Reduced Risk of Scope Creep 5. Higher User Satisfaction • Conclusion • Frequently Asked Questions - o Why Is an SRS Important for E-Learning Platforms? - o What Are the Components of an SRS for E-Learning Platforms? - o What Challenges Are Involved in Developing an SRS for E-Learning Platforms? Importance of SRS for E-Learning Platform A strong SRS for an e-learning platform plays a vital role in the software development process. It acts as a contract between stakeholders and the development team, clearly defining expectations and deliverables. The SRS helps in: • Ensuring all requirements are met. • Providing a clear roadmap for developers. • Reducing misunderstandings and miscommunications. • Facilitating accurate project estimation and planning. • Serving as a reference for validation and verification. Components of an SRS for E-Learning Platform An SRS document for an e-learning platform typically includes several key sections. 1. System Overview The system overview provides a high-level description of the e-learning platform, including its purpose, scope, and objectives.
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This section establishes the foundation for the detailed requirements that follow.

2. Functional Requirements

The functional requirements specify what the e-learning platform should do. These may include:

- User Registration and Login: Allowing students, instructors, and administrators to create accounts and securely access the platform.
- Course Management: Enabling instructors to create, upload, edit, and manage course materials.
- Enrollment Management: Allowing learners to browse and enroll in courses.
- Assignment and Quiz Management: Supporting online assessments, submissions, and grading.
- Video Lectures and Content Delivery: Providing access to recorded lectures, presentations, and study materials.
- Progress Tracking: Monitoring learner performance and course completion.
- Payment and Subscription Management: Handling course purchases, subscriptions, and payment transactions.
- Certification Management: Generating certificates upon successful course completion.
- Discussion Forums and Messaging: Facilitating communication between instructors and learners.

3. Non-Functional Requirements

Non-functional requirements define how the system performs its functions. These include:

- Performance: The platform should support multiple concurrent users without performance degradation.
- Security: User data and payment information should be securely stored and protected.
- Usability: The platform should provide an intuitive and user-friendly interface.
- Reliability: The system should maintain high availability and ensure data consistency.
- Scalability: The platform should support increasing numbers of users and courses.
- Accessibility: The system should be accessible across various devices and comply with accessibility standards.

4. User Interfaces

This section describes the design and layout of the user interfaces. It includes wireframes and mockups for dashboards, course pages, quizzes, discussion forums, and administrative panels.



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5. System Architecture

The system architecture outlines the overall structure of the e-learning platform, including:

- Front-end components.
- Back-end services.
- Database architecture.
- Cloud infrastructure.
- Network architecture.
- Data flow diagrams.

6. Data Requirements

Data requirements specify the types and formats of data handled by the system, including:

- User profiles.
- Course information.
- Learning materials.
- Assessment records.
- Payment details.
- Certificates.
- Activity logs and analytics.

7. External Interface Requirements

These requirements specify how the platform interacts with external systems, such as:

- Payment gateways.
- Video conferencing tools.
- Email and notification services.
- Learning Management System (LMS) integrations.
- Cloud storage services.
- Third-party authentication services.

8. System Constraints

This section identifies limitations or restrictions affecting the system's design and implementation, including:

- Budget constraints.
- Internet connectivity requirements.
- Device compatibility limitations.
- Data privacy regulations.
- Storage and bandwidth limitations.

Developing an SRS for E-Learning Platform

Creating a comprehensive SRS involves several important steps.

1. Requirement Gathering

Requirement gathering is the first step in developing an SRS. This involves collecting information from stakeholders, including:



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- Students.
- Instructors.
- Administrators.
- Educational institutions.
- Technical teams.

Techniques such as interviews, questionnaires, surveys, and observations can be used to gather requirements.

2. Requirement Analysis

Once the requirements are collected, they must be analyzed to ensure feasibility and alignment with project objectives. This process includes:

- Prioritizing requirements.
- Resolving conflicts.
- Identifying missing functionalities.
- Evaluating technical constraints.

3. Requirement Specification

The specification phase involves documenting requirements clearly and systematically.

This includes:

- Functional requirements.
- Non-functional requirements.
- User interface requirements.
- Data requirements.
- External interface requirements.
- System constraints.

4. Requirement Validation

Requirement validation ensures that the documented requirements accurately reflect stakeholder expectations. Validation techniques include:

- Requirement reviews.
- Walkthroughs.
- Inspections.
- Stakeholder approvals.

5. Requirement Management

Requirement management is an ongoing process that tracks and manages changes throughout the project lifecycle. It helps maintain alignment between the platform's functionality and stakeholder needs.

Challenges in Developing an SRS for E-Learning Platform

Developing an SRS for an e-learning platform presents several challenges.



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1. Diverse Learning Requirements

Different learners and instructors may have varying educational needs and preferences. Accommodating these requirements can increase system complexity.

2. Data Privacy and Security

The platform must protect sensitive user information and comply with applicable data protection regulations.

3. Integration with Third-Party Services

Modern e-learning platforms often rely on third-party services such as payment gateways, video conferencing applications, and cloud-based tools. Seamless integration requires careful planning.

4. User Adoption and Engagement

Ensuring that learners and instructors actively use the platform requires an intuitive design, engaging features, and adequate support resources.

Benefits of a Well-Defined SRS for E-Learning Platform

A well-defined SRS offers numerous advantages.

1. Improved Communication

An SRS serves as an effective communication tool between stakeholders and the development team by clearly documenting project requirements and expectations.

2. Enhanced Project Management

A detailed SRS supports accurate planning, scheduling, budgeting, and resource allocation, reducing the likelihood of delays and cost overruns.

3. Better Quality Assurance

The SRS provides a benchmark for testing and validation, ensuring that the developed system satisfies all specified requirements.

4. Reduced Risk of Scope Creep

Clearly defined requirements help manage changes effectively and minimize unnecessary additions during development.

5. Higher User Satisfaction

A well-documented SRS ensures that the final platform meets user expectations, resulting in improved learning experiences and higher adoption rates.

Conclusion

In conclusion, a comprehensive SRS for an e-learning platform is essential for ensuring that the final product meets educational, operational, and technical requirements. By clearly defining

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	system functionalities and specifications, an SRS supports accurate project planning, minimizes development risks, and improves overall software quality. Developing a detailed SRS involves requirement gathering, analysis, specification, validation, and management. Although challenges exist, the benefits of a well-defined SRS are substantial, including improved communication, enhanced project management, better quality assurance, and greater user satisfaction. Frequently Asked Questions 1. Why Is an SRS Important for E-Learning Platforms? An SRS provides a clear understanding of the platform's objectives, requirements, and functionalities, helping ensure successful software development and stakeholder satisfaction. 2. What Are the Components of an SRS for E-Learning Platforms? The key components include system overview, functional requirements, non-functional requirements, user interfaces, system architecture, data requirements, external interface requirements, and system constraints. 3. What Challenges Are Involved in Developing an SRS for E-Learning Platforms? Common challenges include addressing diverse user requirements, ensuring data security, integrating third-party services, and encouraging user adoption and engagement.
Link of student Github profile where lab assignment has been uploaded	https://github.com/TanushriSawalikar/Software-Engineering-Practicals
Conclusion	I have Successfully studied and created a Software requirement Specification document for the E-Learning Platform
Date	13/07/2026